

Lesson Overview

Subject: Computer Science / Technology

Course: Coding for the Web

Grade Level: 7th Grade (MYP Year 2)

Lesson Title: Introduction to Variables

Duration: 60 minutes

Date: 2/06/2026

Learning Objectives

By the end of this lesson, students will be able to:

- Explain what a variable is in computer science using their own words.
 - Identify and use two basic variable types: **number** and **string**.
 - Create a dynamic story using variables.
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IB Alignment (MYP)

Key Concept: Systems

Related Concepts: Logic

Global Context: Scientific and technical innovation

Approaches to Learning:

- Thinking skills (logical reasoning)
- Communication skills (interpreting outputs, debugging)
- Self-management skills (following routines, persistence)

Assessment Criteria:

- Knowing & Understanding: Correct use of variables
 - Applying & Thinking: Variables used meaningfully in a program
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Materials & Resources

- Laptop or Chromebook for each student
- Internet access
- Projector / teacher display
- Shared demo links

Links:

- Warm-up CodePen: <https://codepen.io/mdixon/designs/pen/yJOXqX>
 - Guided practice starter template:
<https://codepen.io/mdixon/designs/pen/xbOaryE?editors=0011>
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Classroom Expectations & Routines (5 min)

Tech Norms:

- Lids at 45° when the teacher is speaking.
- Screens visible and sound off when working.

Coding Norms:

- Errors are good for you.
- Please break your code and read the error message before asking for help.

Collaboration Structure:

- Students may work independently or with a partner.
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Warm-Up (15 minutes)

Teacher Actions:

- Brief introduction to CodePen.
- Open the warm-up CodePen. Scroll through the JS panel. Note the variable section at the top.

Student Task:

- Locate the variable `$ballR`
- Change its value from `10` to `100`
- Challenges:
 - Change the color of the ball / paddles / score
 - Change the dimensions of the game
- **Bonus:** Remove the `//` from line 140.
`// $ballColor = "hsl(" + $ballX + ", 80%, 50%)" ;`

Discussion Prompt:

- How could we make the game harder? Easier?

Formative Check:

- Teacher circulates and confirms students can identify and modify a variable.
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Direct Instruction (10 minutes)

Concept Focus: What is a variable?

Teaching Points:

- Variables as labeled containers (box metaphor)
- Variables store information for later use
- Two variable types today:
 - **Number** (math, counting, scores)
 - **String** (text, names, messages)

Real-World Connections:

What variable *type* would you use to store the following examples?

- Football player's total rushing yards.

- Book title in a library database.
- Price of an item in your online shopping cart.
- What other examples can you think of?

Live Demo:

- Demonstrate `console.log()` message using a hardcoded string: `"Hello, world."`
 - Introduce syntax for a variable. `let variableName = "Dolly";`
 - Print variable using `console.log()`
 - Concatenate a variable with a string.
 - Follow the same steps to add another variable to store a number and concatenate it to the string.
 - **Bonus:** Add an emoji the same way! 🕶️✨💀🌈🔥
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Guided Practice (25 minutes)

Task: Build a dynamic short story program.

Structure:

- Teacher codes live step-by-step
- Students follow along using the starter template

Key Steps:

- Disable CodePen auto-update. Demonstrate the new 'run' button.
- Print a short story (two sentences) using `console.log()`.
- Declare a string variable.
- Declare a number variable.
- Include variables inside `console.log()` stories.

Support:

- Teacher circulates
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Assessment / Evidence of Learning

Primary Evidence:

- Working code using at least:
 - One string variable
 - One number variable

Assessment Method:

- Teacher observation
- Review of student CodePen link or shared file

Success Criteria (Shared with Students):

- Code runs without errors.
 - Variables are clearly named.
 - Output changes when variable values change.
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Extension Activities (Early Finishers)

“Mad Lib” Style Game

- Use multiple string variables
 - Create a “Mad Lib” style game with 5+ variables.
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Wrap-Up / Closure (5 minutes)

Reflection Question (Exit Ticket – written):

- “In your own words, what is a variable and why is it useful?”

Review:

- Revisit lesson objectives.
- Highlight examples of student work.

Preview of Next Lesson:

- Introduction to **math operators** (+, -, *, /)